



FinCast: A Simple Agentic Framework for Financial Time Series Forecasting

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Abstract

Financial time-series forecasting remains challenging because historical prices provide only partial evidence: market movements also respond to information shocks conveyed through earnings disclosures, analyst revisions, and financial news. Classical ARIMA, GARCH, and ARIMAX model families capture non-stationarity, mean dynamics, and volatility, but provide limited guidance when their signals disagree and cannot reason over unstructured textual evidence. We propose **FinCast**, a training-free agentic framework that treats forecasting as evidence-based decision making rather than direct price generation. Given statistical forecasts, retrieved historical analogues, and time-aligned financial news, an LLM evaluates model consensus, case evidence, and news sentiment to revise a cluster-weighted reference trajectory: aligned evidence triggers structural correction, while conflicting evidence shrinks the forecast toward the reference. A reflection step enforces return bounds and audits information leakage. Across NVDA and NFLX over 2010–2020, FinCast lowers macro-averaged RMSE by **10.2%** relative to the strongest baseline and improves directional accuracy by **16.5** percentage points over the baseline average. The implementation is available at <https://github.com/akatsuky999/FinCast>.

Introduction

A central difficulty in financial forecasting is that the most informative signals often do not live in a single modality. Historical prices encode autocorrelation, volatility clustering, and regime-dependent statistical structure, while market-moving events appear as unstructured text in earnings reports, analyst revisions, product announcements, and sector-level news. Prior work has shown that media tone, financial disclosures, and event text carry predictive information for future returns and price movements (Tetlock 2007;

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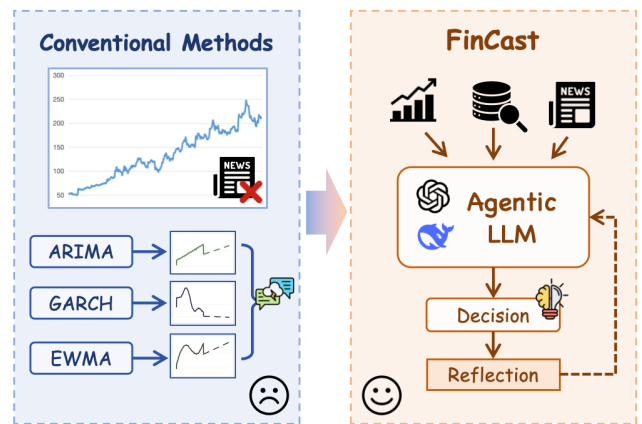


Figure 1: Comparison between conventional numerical forecasting and FinCast, which combines statistical signals, historical analogues, and market news.

Loughran and McDonald 2011; Ding et al. 2015; Xu and Cohen 2018). Classical econometric models have been highly effective at formalizing the numerical side of this problem. However, these models still operate under a static numerical forecasting paradigm. They provide limited support for reconciling disagreement across model families, and no direct mechanism for reasoning over semantic events that may overturn historical extrapolation. Figure 1 illustrates this gap between conventional numerical forecasting pipelines and the broader evidence integration required in practice.

This limitation is especially visible when different sources of evidence point in different directions. An ARIMA model may extrapolate a rising trend, a GARCH model may signal elevated risk, and an ARIMAX model may flatten the trajectory after incorporating exogenous variables. A financial analyst would not average these outputs blindly. Instead, they would inspect the relevant news, compare the current market state with historical analogues, and judge whether the semantic evidence is strong enough to justify a structural adjustment to the forecast.

The challenge becomes more pronounced in long-horizon trajectory forecasting. Classical models often provide dis-

ciplined but conservative numerical paths, especially when trend, seasonality, or exogenous signals are weak. Such forecasts are valuable as no-leakage anchors, but they are not designed to decide when semantic events or historical analogues justify a stronger departure from the reference trajectory. This motivates a framework that treats statistical forecasts as evidence to be inspected and adjusted, rather than as the final answer.

Existing learning-based methods only partially address this gap. Multimodal deep models can fuse prices and text and often improve stock movement prediction (Jin et al. 2024; Liu et al. 2025). However, these methods are usually trained or prompted as fixed input-output predictors and often capture correlations rather than event-level causal implications. Recent time-series foundation models improve zero-shot forecasting, but they still act as black-box generators: they do not actively compare hypotheses, inspect external evidence, revise forecasts, or audit potential failures.

We propose **FinCast**, a training-free agentic framework that recasts financial forecasting as multi-source evidence aggregation rather than direct price generation. Instead of asking the LLM to forecast from scratch, FinCast provides a structured brief containing statistical forecasts, retrieved historical analogues, and time-aligned financial news. The agent reasons over agreement and conflict among these sources to adjust a conservative reference trajectory: aligned evidence supports a structural correction, while conflicting evidence pulls the prediction back toward the anchor. A reflection module then checks return bounds and potential information leakage, revising the forecast when needed. This closed-loop design combines statistical discipline, retrieval-based anchoring, and semantic reasoning without task-specific training.

We evaluate FinCast on two Nasdaq equities with distinct market regimes, NVDA and NFLX, from 2010–2020. FinCast reduces macro-averaged RMSE by **10.2%** over the strongest baseline, achieves **59.5%** directional accuracy compared with a **43.0%** baseline average, and is the only method that remains among the top performers across both assets. Ablation studies show that historical case retrieval and LLM-based semantic reasoning provide the dominant gains, indicating that statistical ensemble signals and event-level textual interpretation are complementary.

Our contributions are threefold:

- We formulate financial time-series forecasting as a multi-source evidence-based decision problem, exposing the limitations of static numerical regression in handling model disagreement and semantic shocks.
- We propose **FinCast**, a training-free agentic framework that uses consistency among statistical forecasts, retrieved historical cases, and financial news to guide structural trajectory correction.
- We validate FinCast on two contrasting Nasdaq stocks, showing improved numerical and directional accuracy, with ablations confirming the contribution and complementarity of its core components.

Related Work

Classical financial forecasting. Financial forecasting has long been dominated by parsimonious statistical models because they are data-efficient, interpretable, and robust under limited supervision. Our baselines follow this tradition, spanning persistence and mean-based rules, autoregressive return models, ARIMA and ARIMAX for non-stationary dynamics and exogenous effects (Hyndman and Khandakar 2008; Pankratz 1991), exponential-smoothing families including AutoETS and Theta (Hyndman et al. 2002; Assimakopoulos and Nikolopoulos 2000; Hyndman and Billah 2003), and ARCH and GARCH style volatility models (Engle 1982; Bollerslev 1986). These models align well with financial stylized facts such as weak-form efficiency and volatility clustering (Fama 1970), but they remain fundamentally numerical: they cannot interpret event text or decide when disagreement across model families reflects a true regime shift.

Deep learning based forecasting. Deep forecasting moved from RNN-based and TCN-based sequence models such as DeepAR, LSTNet, and ModernTCN (Salinas et al. 2020; Lai et al. 2018; Bai, Kolter, and Koltun 2018; Luo and Wang 2024) to Transformer-based forecasters, which now dominate the literature through architectures such as Informer, Autoformer, PatchTST, DLinear, TimesNet, and iTransformer (Zhou et al. 2021; Wu et al. 2021; Nie et al. 2023; Zeng et al. 2023; Wu et al. 2023; Liu et al. 2024b). Recent work further explores pretrained and multimodal forecasting, including Chronos, AutoTimes, Time-LLM, TimeCMA, Time-VLM, Time-MMD, and Aurora (Ansari et al. 2024; Liu et al. 2024c; Jin et al. 2024; Liu et al. 2025; Zhong et al. 2025; Liu et al. 2024a; Wu et al. 2026). This trend supports our core intuition that language and other auxiliary modalities can improve forecasting, but most existing models still learn a direct mapping from context to forecast rather than explicitly reasoning over conflicting statistical and textual evidence.

Agentic time-series forecasting. A more recent line treats forecasting as iterative reasoning rather than one-shot prediction. Built on chain-of-thought, tool use, and reflection (Wei et al. 2022; Yao et al. 2023; Shinn et al. 2023), methods such as AlphaCast, TimeCopilot, and Time-R1 introduce multi-step deliberation into time-series forecasting (Zhang et al. 2025; Garza and Rosillo Garcia 2025; Luo et al. 2026). Most existing agentic evaluations, however, focus on generic forecasting benchmarks or domains such as electricity, traffic, and weather, where seasonality and repeated temporal structure are relatively strong. Financial forecasting is a harder setting: the signal-to-noise ratio is lower, regimes change more abruptly, and textual events can overturn otherwise plausible numerical trends. FinCast adapts this agentic view to finance by grounding forecasting in cross-source evidence rather than purely numerical prediction.

Problem Formulation

The task is multi-source financial trajectory forecasting. At forecast origin t , a forecaster observes a lookback win-

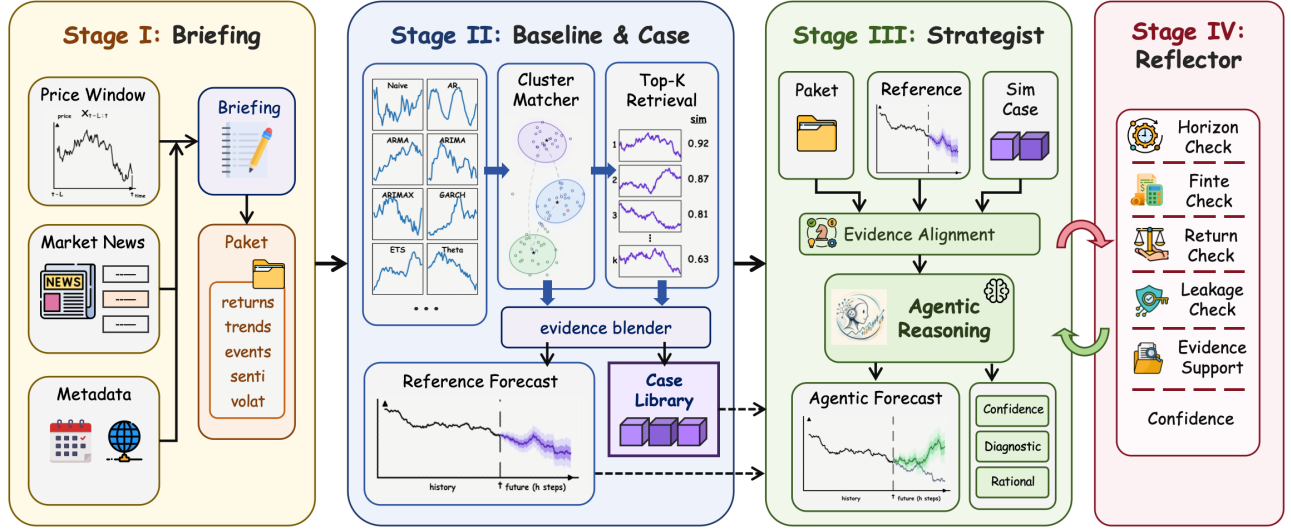


Figure 2: Overview of FinCast. Briefing prepares market context, Baseline produces statistical forecasts and retrieved analogues, Strategist synthesizes cross-source evidence into a trajectory forecast, and Reflector validates the result before acceptance or revision.

dow containing historical prices $\mathbf{X}_{t-L:t}$, aligned exogenous variables $\mathbf{E}_{t-L+1:t}$, and market news $\mathcal{N}_{t-L:t}$, and predicts the future H -day price path $\mathbf{Y}_{t+1:t+H}$. Let $\mathcal{F}_t = \{\mathbf{X}_{t-L:t}, \mathbf{E}_{t-L+1:t}, \mathcal{N}_{t-L:t}\}$ denote all information available at decision time. The objective is to construct a decision rule

$$\hat{\mathbf{Y}}_{t+1:t+H} = f(\mathcal{F}_t),$$

that minimizes trajectory error $\mathcal{L}(\hat{\mathbf{Y}}_{t+1:t+H}, \mathbf{Y}_{t+1:t+H})$ while retaining directional fidelity. All methods are subject to a strict no-leakage constraint: no future prices, exogenous variables, or post- t news may be used. This formulation covers purely numerical baselines that use subsets of $\mathcal{F}_t$, as well as multimodal or agentic forecasters that transform the same admissible evidence before making a decision.

Methodology

FinCast separates *evidence construction* from *forecasting decisions*. Rather than using an LLM as a direct price generator, it first constructs a disciplined evidence set from market history, retrieved analogues, and time-aligned news. The agent then judges whether the evidence warrants a controlled departure from a conservative reference trajectory:

$$\hat{\mathbf{Y}}_{t+1:t+H} = \mathcal{R}(\mathcal{S}(\mathbf{X}_{t-L:t}, \mathcal{N}_{t-L:t}, \mathcal{C}_t, \mathcal{B}_t)),$$

where $\mathcal{S}$ and $\mathcal{R}$ denote the *Strategist* and *Reflector*, while $\mathcal{C}_t$ and $\mathcal{B}_t$ are constructed by the upstream *Briefing* and *Baseline* stages. This design anchors the forecast in observable financial evidence while allowing strong semantic events to reshape the trajectory. Figure 2 summarizes the workflow.

Briefing Stage

The Briefing stage converts raw observations into a compact financial state without issuing a forecast. Given prices $\mathbf{X}_{t-L:t}$, exogenous variables $\mathbf{E}_{t-L+1:t}$, and news $\mathcal{N}_{t-L:t}$, it

computes log returns $r_\tau = \log X_\tau - \log X_{\tau-1}$ and summarizes the window through trend, realized volatility, draw-down, activity, and tail risk:

$$\phi_t = [\bar{r}_t, \sigma_t, \text{MDD}_t, q_{0.95}(|r|), q_{0.99}(|r|)],$$

where σ_t is realized volatility and MDD_t is maximum draw-down over the lookback window. Recent headlines are bucketed by time and summarized only as information available before t . The briefing packet $\mathcal{P}_t = (\phi_t, \mathbf{E}_{t-L+1:t}, \mathcal{N}_{t-L:t})$ therefore describes market state and risk without generating prices.

Baseline Stage

The Baseline stage produces the numerical evidence consumed by the downstream agent. Twelve classical forecasters are run independently, spanning naive rules, autoregressive models, ARIMA, ARIMAX, GARCH-style models, exponential smoothing, and random-walk variants. Formal specifications, the k -medoids routine, and reference construction are given in Appendix B. Beyond serving as competitors, these models define a diverse candidate set

$$\mathcal{B}_t = \left\{ \hat{\mathbf{Y}}_{t+1:t+H}^{(m)} \right\}_{m=1}^M,$$

from which model disagreement, consensus, and a conservative reference path are estimated.

Baseline evidence is further grounded by a historical case library. Each case i stores a normalized lookback trajectory, its realized future path, and model losses under the same horizon. At inference time, FinCast scores cases with the same composite similarity used by the retrieval module:

$s(t, i) = \lambda_1 s_{\text{shape}} + \lambda_2 s_{\text{dtw}} + \lambda_3 s_{\text{trend}} + \lambda_4 s_{\text{vol}} + \lambda_5 s_{\text{acf}}$, which compares price shape, local temporal warping, trend, volatility, and autocorrelation. The top cases form $\mathcal{C}_t$, with

retrieval weights $a_i \propto [2 - s(t, i)]^{-1}$. Their historical model losses induce local model weights, while the same library is clustered offline so that recurrent regimes yield stable model-weight patterns.

The two views are combined into a single reference trajectory. Let $v_t^{(m)} \propto \sum_{i \in \mathcal{C}_t} a_i / (\ell_i^{(m)} + \epsilon)$ be the case-induced model weight obtained from historical losses, and let $w_t^{(m)}$ be the cluster-induced model weight. These define

$$\mathbf{Y}_{\text{case}} = \sum_{m=1}^M v_t^{(m)} \hat{\mathbf{Y}}_{t+1:t+H}^{(m)},$$

$$\mathbf{Y}_{\text{cluster}} = \sum_{m=1}^M w_t^{(m)} \hat{\mathbf{Y}}_{t+1:t+H}^{(m)}.$$

FinCast uses

$$\mathbf{Y}_{\text{ref}} = \beta \mathbf{Y}_{\text{cluster}} + (1 - \beta) \mathbf{Y}_{\text{case}},$$

where β increases when the current window confidently matches a known regime. The stage returns $\mathcal{B}_t$ and the grounded reference trajectory $\mathbf{Y}_{\text{ref}}$.

Strategist Stage

The Strategist is the core agentic component. It receives the Briefing and Baseline packets, then decides whether the reference forecast should be trusted, adjusted, or left nearly unchanged. The LLM acts as a *decision-maker*: it reasons over precomputed evidence rather than serving as an unconstrained price generator.

The decision is organized around three directional signals: model consensus s_{model} , retrieved case evidence s_{case} , and news evidence s_{news} . The Strategist evaluates evidence alignment and uncertainty,

$$A_t = \left\lfloor \frac{s_{\text{model}} + s_{\text{case}} + s_{\text{news}}}{3} \right\rfloor, \quad U_t = \text{Disp}(\mathcal{B}_t) + \sigma_t,$$

where A_t is high when the evidence points in a common direction, and U_t grows with model disagreement and realized volatility. In LLM mode, these quantities are provided as structured diagnostics and the agent emits a price path; in deterministic fallback mode they reduce to a direction-strength adjustment:

$$\tilde{\mathbf{Y}}_{t+1:t+H} = \mathbf{Y}_{\text{ref}} \odot \exp(\gamma_t \mathbf{r}), \quad \gamma_t \propto \frac{A_t}{1 + U_t},$$

where $\mathbf{r}$ is a monotone horizon ramp. Aligned model, case, and news evidence permits a stronger correction; weak or contradictory evidence shrinks γ_t toward zero, keeping the forecast close to $\mathbf{Y}_{\text{ref}}$. If the LLM is unavailable, FinCast falls back to this deterministic rule with the same conservative bias.

Reflector Stage

The Reflector is a deterministic validation layer applied after the Strategist proposes $\tilde{\mathbf{Y}}_{t+1:t+H}$. It checks horizon length, timestamp alignment, positivity, finite values, leakage, and evidential support for material deviations. It also clips step

returns with the empirical bound used in the implementation:

$$\left| \log \tilde{Y}_{t+h} - \log \tilde{Y}_{t+h-1} \right| \leq B_t^{\max}, \quad h = 1, \dots, H.$$

Here $B_t^{\max}$ is computed from high historical return quantiles, realized volatility, and an absolute cap. Forecasts that pass all checks are accepted. Otherwise, violations are returned to the Strategist as structured feedback for a small number of retries. If validation still fails, FinCast falls back to $\mathbf{Y}_{\text{ref}}$. This audit step permits semantic intervention while preserving no-leakage, scale consistency, and financially plausible dynamics.

Experiments

Experimental Setup

We evaluate FinCast on two NASDAQ equities with distinct regimes: NVDA, a trend-driven semiconductor stock that underwent a major valuation re-rating during 2010–2020, and NFLX, a high-volatility internet entertainment stock with stronger mean-reverting behavior. Each series contains 2,539 trading days from 2010-01-06 to 2020-02-06, with OHLCV prices, nine lagged exogenous variables, and time-aligned financial news from analyst ratings and partner media headlines. The task uses a long-horizon forecasting protocol with a 60-day lookback and forecast horizon. Windows are split into 90% training and 10% test sets at the window level with seed 5053, giving 726 training windows and 81 test windows per stock. Training windows are used only to build the case library and K-Medoids regime memory; all reported results are computed on held-out windows.

Table 1: Statistical models used in the Baseline stage.

Category	Models
Naive	LastClose, HistoricAverage, SeasonalNaive
AR and MA	ARReturn, ARMAReturn
ARIMA	ARIMALogPrice
ARIMAX	ARIMAXPrice
GARCH	ARMAGARCHReturn
Exponential	EWMAReturn, AutoETS, Theta
Random Walk	RandomWalkDrift

We compare against the twelve statistical forecasters in Table 1, spanning naive rules, AR and ARMA models, ARIMA and ARIMAX variants, GARCH-style volatility modeling, exponential smoothing, Theta, and random-walk drift. All methods use the same admissible information and predict the same long-horizon trajectory. This setting is demanding for classical forecasters, which often become conservative when reliable trend, seasonality, or exogenous signals are limited. We retain this setting because FinCast uses these forecasts as interpretable numerical anchors rather than unconstrained trajectory generators; substantial departures from the reference path must be supported by retrieved analogues, news evidence, and reflector validation.

We report root mean squared error (RMSE), mean absolute error (MAE), and directional accuracy (DirAcc). For a

Table 2: Test performance on NVDA and NFLX. Macro scores are equally averaged over the two stocks. Ranking markers: **bold** denotes the best value, **red** denotes the second-best value, and underline denotes the third-best value.

Model	Average				NFLX			NVDA		
	RMSE↓	MAE↓	DirAcc↑	Rank↓	RMSE↓	MAE↓	DirAcc↑	RMSE↓	MAE↓	DirAcc↑
ARReturn	80.31	31.39	0.538	10.5	141.62	52.82	0.550	19.00	9.95	0.526
RandomWalkDrift	78.50	30.57	<u>0.571</u>	11.0	136.32	50.42	<u>0.574</u>	20.67	10.71	0.569
ARMAReturn	77.96	30.51	0.554	10.0	136.21	50.76	0.564	19.71	10.25	0.544
EWMAReturn	74.17	32.80	0.490	11.0	124.81	53.94	0.413	23.53	11.67	0.566
Theta	63.90	26.01	0.575	5.0	111.22	43.35	0.613	16.58	8.66	0.536
ARMAGARCHReturn	63.14	28.65	0.483	9.5	104.39	46.35	0.469	21.90	10.94	0.498
ARIMAXPrice	57.50	<u>25.08</u>	0.495	6.5	98.24	41.03	0.521	16.77	9.12	0.468
ARIMALogPrice	57.29	25.28	0.367	5.5	97.81	41.47	0.350	16.76	9.09	0.384
AutoETS	57.14	25.00	0.001	4.5	97.59	40.99	0.000	16.70	9.00	0.001
LastClose	57.14	25.00	0.001	3.5	97.59	40.99	0.000	<u>16.70</u>	<u>9.00</u>	0.001
SeasonalNaive	<u>56.98</u>	25.59	0.394	5.0	<u>96.79</u>	42.10	0.376	17.16	9.08	0.412
HistoricAverage	56.62	31.97	0.434	7.5	88.56	50.22	0.441	24.67	13.73	0.427
FinCast	50.85	24.95	0.595	1.5	85.10	<u>41.05</u>	0.627	16.60	8.85	<u>0.564</u>

test set with N forecast origins and horizon H , RMSE and MAE are

$$\text{RMSE} = \sqrt{\frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H \left(\hat{Y}_{i,h} - Y_{i,h} \right)^2},$$

$$\text{MAE} = \frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H \left| \hat{Y}_{i,h} - Y_{i,h} \right|.$$

Metrics are computed per stock and macro-averaged with equal weight. Lower RMSE, MAE, and rank are better, while higher DirAcc indicates stronger directional fidelity. The Strategist uses DeepSeek-V4-Flash. For each stock, the case memory contains 807 precomputed windows and the regime memory uses $k = 6$ K-Medoids clusters.

Main Results

Table 2 reports results for the macro average and the two individual stocks. FinCast obtains the best macro RMSE (50.85), improving over the strongest baseline, HistoricAverage (56.62), by 10.2%. It also achieves the best macro MAE and DirAcc. The directional improvement is substantial: FinCast reaches 59.5% DirAcc, compared with a 43.0% baseline average. Its average rank is 1.5, well below all baselines, indicating that the gain is not driven by one favorable asset.

The per-stock results illustrate the brittleness of fixed statistical rules. Theta is strongest by RMSE on NVDA but drops sharply on NFLX; HistoricAverage is competitive on NFLX but ranks last on NVDA. This reversal matches the underlying regimes: persistent trend favors momentum-like extrapolation on NVDA, whereas mean reversion makes value anchoring more effective on NFLX. FinCast is the only method that remains in the top two by RMSE on both stocks, since its final decision is conditioned on model

consensus, retrieved analogues, and contemporaneous news rather than a single invariant formula.

Directional accuracy shows a complementary advantage. On NFLX, FinCast is the only method above 60% DirAcc, reaching 62.7%. On NVDA, simple directional extrapolators can match or slightly exceed its DirAcc, but at much larger trajectory error; RandomWalkDrift has the best NVDA DirAcc but 24.5% higher RMSE than FinCast. Thus FinCast does not simply optimize signs. It preserves path scale while using semantic and historical evidence to improve directional decisions.

On NFLX, Diebold–Mariano tests over the 81 test windows show significant improvements over HistoricAverage, EWMAReturn, and ARMAGARCHReturn at $p < 0.05$. The DM statistic is negative against all twelve baselines, indicating consistently lower forecast loss. Given the 81-window test set, we interpret non-significant pairwise comparisons conservatively rather than as evidence of no difference.

Ablation Study

We isolate component contributions on NFLX, where regime changes and event-driven movements make forecasting more difficult. The ablation starts from the reference trajectory produced by statistical model weighting and then adds deterministic adjustment, neighborhood anchoring, mean shrinkage, and the LLM Strategist.

Neighbor anchoring is the largest single contributor, accounting for 49.0% of the total RMSE reduction. The unanchored reference is intentionally conservative but can underreact over long horizons. Calibrating trajectory magnitude with similar historical windows makes the reference path more responsive to the current regime.

The LLM agent provides the second-largest gain and changes the form of the correction. The deterministic for-

Table 3: Ablation on NFLX test windows. Share denotes each component’s fraction of the total RMSE reduction from configuration A to E.

Configuration	RMSE↓	Δ	Share	DirAcc↑
A: Reference only	120.24	–	–	0.083
B: + Formula	115.64	-4.60	13.1%	0.083
C: + Neighbor anchor	98.42	-17.22	49.0%	–
D: + Mean shrinkage	96.08	-2.34	6.7%	0.521
E: + LLM agent	85.10	-10.98	31.2%	0.627

mula can only apply a scalar direction-strength adjustment, whereas the Strategist reads headline text, compares retrieved future paths with the statistical reference, and produces non-uniform trajectory adjustments when semantic and historical evidence support them. This reduces RMSE by an additional 10.98 points and raises DirAcc from 52.1% to 62.7%.

The components are complementary. Neighbor anchoring gives the agent a sharper and better-scaled reference, while the LLM reduces over-reliance on historical similarity when news suggests a regime shift. The improvement therefore comes from evidence coordination rather than from simply adding another predictor to the ensemble.

Interpretation of Baselines

The baseline ordering reflects concrete regime-dependent trade-offs. On NVDA, where the price path is more persistent, Theta gives the lowest RMSE and simple persistence models remain competitive in scale error; however, the best directional score comes from RandomWalkDrift at the cost of substantially larger RMSE. On NFLX, HistoricAverage is the strongest RMSE baseline, consistent with a more mean-reverting series, while LastClose and AutoETS obtain low MAE by staying close to the current level but fail almost completely on directional accuracy. AutoETS matches LastClose because its selected smoothing specification becomes a level-only forecast on these windows. ARMAGARCHReturn remains in the middle range on both stocks, which is expected for a model designed primarily for conditional variance rather than point-path accuracy. These patterns show that each baseline encodes a useful but narrow assumption; FinCast improves by treating such assumptions as conservative evidence and adjusting them with regime, case, and news information.

Discussion

The results suggest that FinCast improves forecasting not by replacing statistical models, but by changing how their signals are used. The baseline ensemble already contains useful information from twelve classical forecasting paradigms, including persistence, autoregression, smoothing, volatility modeling, and random-walk drift. However, these models are tied to narrow inductive assumptions and become brittle when the long-horizon trajectory depends on events, regime shifts, or conflicting evidence. FinCast treats the baseline output as an interpretable numerical anchor and

asks the LLM agent to revise it only when external evidence supports doing so. This design adds two capabilities that purely numerical baselines lack: semantic reading of financial news, which separates material firm-level events from generic market sentiment, and structural reasoning over trajectory shapes, which allows retrieved historical analogues to inform the forecast without being copied mechanically. The ablation supports this interpretation: neighbor anchoring provides the largest scale correction, while the LLM Strategist contributes the second-largest gain and accounts for 31.2% of the total RMSE reduction. Directional accuracy further clarifies the benefit. In financial settings, the sign of future movement is often more actionable than exact point magnitude for allocation, hedging, and scenario analysis. FinCast improves macro directional accuracy to 59.5%, 16.5 points above the baseline average, while also reducing RMSE and MAE. This indicates that the agent does not merely make more directional bets; it improves direction while preserving plausible price scale through case evidence and Reflector validation.

Limitations

FinCast is intended to enhance classical financial time-series pipelines, not to replace specialized deep sequence forecasters. We therefore restrict the baseline library to interpretable statistical models to isolate whether agentic reasoning improves the same conventional forecasting foundation. This makes the comparison fair for our research question, but also means the results should not be interpreted as state-of-the-art performance against large neural forecasting systems. The value of FinCast is complementary: it shows that event interpretation, historical analogue reasoning, and multi-source evidence synthesis can improve transparent numerical anchors in settings where classical models are weak.

The empirical scope is also limited by API cost and inference time. We evaluate only on NVDA and NFLX from 2010–2020; both are large-cap, technology-related NASDAQ stocks with dense news coverage and growth-style price dynamics. Broader tests across sectors, market-cap ranges, and lower-news regimes are needed to assess generality.

Finally, we keep the LLM backbone fixed. We do not run model-family ablations across larger proprietary models, smaller open-source models, or domain-specialized financial LLMs. Future work should examine how much of the gain comes from the workflow itself versus the reasoning capacity of the selected model, and whether lower-cost agents can preserve the same evidence-grounded behavior.

Conclusion

We introduced FinCast, an agentic framework that augments classical financial time-series models with evidence-grounded LLM reasoning. By combining statistical anchoring, historical analogue retrieval, news interpretation, and deterministic validation, FinCast preserves the discipline of traditional models while allowing forecasts to adapt to events and regime evidence. Experiments on NVDA and

NFLX show consistent gains over twelve statistical baselines, including a 10.2% macro RMSE reduction over the strongest baseline and 59.5% directional accuracy. The ablation further shows that case anchoring and LLM reasoning are complementary: retrieved analogues improve scale calibration, while the Strategist revises trajectories when model, news, and case evidence jointly support a correction. These results suggest a practical hybrid forecasting paradigm in which statistical models provide transparent priors, language agents synthesize external evidence, and rule-based validators enforce plausibility and no-leakage constraints.

References

- Ansari, A. F.; Stella, L.; Turkmen, A. C.; Zhang, X.; Mercado, P.; Shen, H.; Shchur, O.; Rangapuram, S. S.; Arango, S. P.; Kapoor, S.; Zschiegner, J.; Maddix, D. C.; Wang, H.; Mahoney, M. W.; Torkkola, K.; Wilson, A. G.; Bohlke-Schneider, M.; and Wang, B. 2024. Chronos: Learning the Language of Time Series. *Transactions on Machine Learning Research*.
- Assimakopoulos, V.; and Nikolopoulos, K. 2000. The Theta Model: A Decomposition Approach to Forecasting. *International Journal of Forecasting*, 16(4): 521–530.
- Bai, S.; Kolter, J. Z.; and Koltun, V. 2018. An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. arXiv:1803.01271.
- Bollerslev, T. 1986. Generalized Autoregressive Conditional Heteroskedasticity. *Journal of Econometrics*.
- Ding, X.; Zhang, Y.; Liu, T.; and Duan, J. 2015. Deep Learning for Event-Driven Stock Prediction. In *Proceedings of the International Joint Conference on Artificial Intelligence*.
- Engle, R. F. 1982. Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. *Econometrica*, 50(4): 987–1007.
- Fama, E. F. 1970. Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance*, 25(2): 383–417.
- Garza, A.; and Rosillo Garcia, R. 2025. TimeCopilot. NeurIPS 2025 Workshop on Better Time Series Forecasting with Foundation Models and LLMs.
- Hyndman, R. J.; and Billah, B. 2003. Unmasking the Theta Method. *International Journal of Forecasting*, 19(2): 287–290.
- Hyndman, R. J.; and Khandakar, Y. 2008. Automatic Time Series Forecasting: The forecast Package for R. *Journal of Statistical Software*.
- Hyndman, R. J.; Koehler, A. B.; Snyder, R. D.; and Grose, S. 2002. A State Space Framework for Automatic Forecasting Using Exponential Smoothing Methods. *International Journal of Forecasting*, 18(3): 439–454.
- Jin, M.; Wang, S.; Ma, L.; Chu, Z.; Zhang, J. Y.; Shi, X.; Chen, P.-Y.; Liang, Y.; Li, Y.-F.; Pan, S.; and Wen, Q. 2024. Time-LLM: Time Series Forecasting by Reprogramming Large Language Models. In *International Conference on Learning Representations*.
- Lai, G.; Chang, W.-C.; Yang, Y.; and Liu, H. 2018. Modeling Long- and Short-Term Temporal Patterns with Deep Neural Networks. In *The 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*, 95–104.
- Liu, C.; Xu, Q.; Miao, H.; Yang, S.; Zhang, L.; Long, C.; Li, Z.; and Zhao, R. 2025. TimeCMA: Towards LLM-Empowered Multivariate Time Series Forecasting via Cross-Modality Alignment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, 18780–18788.
- Liu, H.; Xu, S.; Zhao, Z.; Kong, L.; Kamarthi, H.; Sasanur, A. B.; Sharma, M.; Cui, J.; Wen, Q.; Zhang, C.; and Prakash, B. A. 2024a. Time-MMD: Multi-Domain Multimodal Dataset for Time Series Analysis. Accepted by NeurIPS 2024 Datasets and Benchmarks Track, arXiv:2406.08627.
- Liu, Y.; Hu, T.; Zhang, H.; Wu, H.; Wang, S.; Ma, L.; and Long, M. 2024b. iTransformer: Inverted Transformers Are Effective for Time Series Forecasting. In *International Conference on Learning Representations*.
- Liu, Y.; Qin, G.; Huang, X.; Wang, J.; and Long, M. 2024c. AutoTimes: Autoregressive Time Series Forecasters via Large Language Models. NeurIPS 2024.
- Loughran, T.; and McDonald, B. 2011. When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks. *Journal of Finance*.
- Luo, D.; and Wang, X. 2024. ModernTCN: A Modern Pure Convolution Structure for General Time Series Analysis. In *International Conference on Learning Representations*.
- Luo, Y.; Zhou, Y.; Cheng, M.; Wang, J.; and Wang, D. 2026. Time Series Forecasting as Reasoning: A Slow-Thinking Approach with Reinforced LLMs. ICLR 2026 submission.
- Nie, Y.; Nguyen, N. H.; Sinthong, P.; and Kalagnanam, J. 2023. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations*.
- Pankratz, A. 1991. *Forecasting with Dynamic Regression Models*. Wiley.
- Salinas, D.; Flunkert, V.; Gasthaus, J.; and Januschowski, T. 2020. DeepAR: Probabilistic Forecasting with Autoregressive Recurrent Networks. *International Journal of Forecasting*, 36(3): 1181–1191.
- Shinn, N.; et al. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems*.
- Tetlock, P. C. 2007. Giving Content to Investor Sentiment: The Role of Media in the Stock Market. *Journal of Finance*.
- Wei, J.; et al. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems*.
- Wu, H.; Hu, T.; Liu, Y.; Zhou, H.; Wang, J.; and Long, M. 2023. TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis. In *International Conference on Learning Representations*.
- Wu, H.; Xu, J.; Wang, J.; and Long, M. 2021. Autoformer: Decomposition Transformers with Auto-Correlation

for Long-Term Series Forecasting. In *Advances in Neural Information Processing Systems*.

Wu, X.; Jin, J.; Qiu, W.; Chen, P.; Shu, Y.; Yang, B.; and Guo, C. 2026. Aurora: Towards Universal Generative Multimodal Time Series Forecasting. In *International Conference on Learning Representations*.

Xu, Y.; and Cohen, S. B. 2018. Stock Movement Prediction from Tweets and Historical Prices. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*.

Yao, S.; et al. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*.

Zeng, A.; Chen, M.; Zhang, L.; and Xu, Q. 2023. Are Transformers Effective for Time Series Forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Zhang, X.; Gao, T.; Cheng, M.; Pan, B.; Guo, Z.; Liu, Y.; Tao, X.; and Liu, Q. 2025. AlphaCast: A Human Wisdom-LLM Intelligence Co-Reasoning Framework for Interactive Time Series Forecasting. arXiv:2511.08947.

Zhong, S.; Ruan, W.; Jin, M.; Li, H.; Wen, Q.; and Liang, Y. 2025. Time-VLM: Exploring Multimodal Vision-Language Models for Augmented Time Series Forecasting. arXiv:2502.04395.

Zhou, H.; Zhang, S.; Peng, J.; Zhang, S.; Li, J.; Xiong, H.; and Zhang, W. 2021. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, 11106–11115.

Appendix A: A Worked Example of LLM Reasoning and Forecast Revision



Purpose. This appendix gives a concrete held-out example showing how the FinCast agent revises a conservative statistical reference forecast. The example is not intended as a new experiment; it is a trace-style case study that exposes the information given to the LLM, the evidence it compares, the revision it makes, and the deterministic checks applied by the Reflector.

Stock	Forecast origin	Horizon	Agent role
NFLX	March 19, 2018	60 trading days	revise reference

The LLM receives a structured research brief rather than raw time series. Its task is to decide whether the reference path should be trusted, amplified, dampened, or reshaped.

A.1 Input Evidence Packet

Numerical anchor and diagnostics.

Reference	\$313.48 → \$343.97; 9.9% monotone gain from the statistical anchor.
Case vote	60% upward versus 40% downward; positive but not decisive.
Model vote	Consensus is UP; mean relative dispersion is 14.73%.
Regime	Volatility 2.66%, drawdown -12.34%, skewness 0.31, and positive trend slope.
Bound	One-step log-return cap is 8.38% from historical extremes.

News packet seen by the LLM.

Usage signal	Customers prefer binge-watching on television, supporting durable engagement.
Event signal	First-quarter earnings will be released on April 16, 2018.
Valuation signal	Streaming leadership is strong, but valuation caution remains.
Market noise	Lyft, Facebook, and broad technology headlines are treated as background rather than direct NFLX evidence.
The lexical pre-score is positive, but materiality is decided at the event level.	

Retrieved neighbor.

The closest historical neighbor has an early rise followed by a severe late decline: approximately \$663.2 → peak \$707.6 near step 16 → end \$97.5. This is an extreme “early peak then collapse” shape. The agent must decide whether to copy this shape, ignore it, or use only its early momentum as qualitative evidence.

A.2 Reasoning and Revision Trace

1

Read the news semantically.

The agent separates firm-specific signals from broad market noise. Netflix’s streaming leadership, TV-based binge watching, and the upcoming earnings announcement are treated as positive or mildly positive firm-level evidence. The Facebook data-driven sell-off is treated as sector pressure rather than direct deterioration in Netflix fundamentals. The Loop Capital headline is mixed: it affirms market leadership while raising valuation caution.

2

Compare neighbor shape with the current regime.

The retrieved neighbor contains an early upward move, but its later collapse is far outside the current window’s drawdown scale. The agent therefore rejects mechanical shape transfer. It keeps the useful qualitative part, early upward momentum in a rising regime, while discarding the implausible late crash.

3

Fuse evidence and choose revision strength.

The evidence is partially aligned: cases and models point upward, news is moderately positive, but case support is only 0.60 and model dispersion is elevated. The agent selects a medium-strength revision rather than an aggressive extrapolation.

4

Generate the corrected trajectory.

The final path preserves the reference’s monotone upward structure, increases the early and middle-horizon slope, and avoids a late reversal. The terminal gain increases from 9.9% in the reference path to 14.3% in the FinCast path, while remaining below historical step-return bounds.

A.3 Reflector Audit

Table A.1: Deterministic audit applied after the LLM proposes its trajectory.

Check	Criterion and observed value	Result
C1	Horizon length is 60; all predicted prices are finite and positive.	PASS
C2	Mean predicted price divided by last close is 1.052, within the allowed range [0.5, 2.0].	PASS
C3	Maximum absolute one-step log return is 0.021, below the empirical bound 0.084.	PASS
C4	No future dates or unsupported numeric claims are introduced.	PASS
C5	Mean deviation from the reference is 2.1%, above the trigger threshold 1.33%, but the deviation is supported by aligned news, case, and model evidence.	PASS
C6	The generated path is not an exact copy of any single statistical baseline.	PASS

A.4 Outcome

Table A.2: Outcome of the NFLX reasoning example.

	Reference	FinCast	Actual
Direction and gain	Up 9.9%	Up 14.3%	Up 19.7%
RMSE	17.0	15.0	–
Relative improvement	–	-11.5%	–

This example shows the intended role of the LLM agent. It does not freely invent a price path. Instead, it reads news at the event level, interprets retrieved trajectories rather than copying them, adjusts a conservative reference only when evidence is sufficiently aligned, and then submits the trajectory to deterministic validation. In this case the final forecast remains conservative relative to the realized move, but it improves over the statistical anchor while preserving scale, leakage, and plausibility constraints.

Appendix B: Method Details

This appendix records the twelve statistical forecasters, k -medoids regime summaries, and the reference trajectory $\mathbf{Y}_{\text{ref}}$.

B.1 Baseline forecasters

Let $P_{t-L+1}, \dots, P_t$ denote lookback closing prices at forecast origin t , with L the lookback length and $T = L$. Define log returns $r_\tau = \log P_\tau - \log P_{\tau-1}$ for $\tau = t - L + 2, \dots, t$. Each forecaster outputs $\hat{\mathbf{P}}_{t+1:t+H} = (\hat{P}_{t+1}, \dots, \hat{P}_{t+H})$. Whenever estimation fails or produces non-finite or implausible values, the forecast is replaced by the constant path at the last observed close P_t (naive level).

Table 3: Twelve statistical baselines: compact role statement and formal definition.

Model	Paradigm	Definition
LastClose	Naive level	$\hat{P}_{t+h} = P_t$ for all $h \in \{1, \dots, H\}$.
HistoricAverage	Mean reversion to window mean	$\bar{P} = \frac{1}{T} \sum_{\tau=t-L+1}^t P_\tau$ and $\hat{P}_{t+h} = \bar{P}$.
SeasonalNaive	Short periodic pattern	Let $s = \min(5, T)$. Tile the suffix $(P_{t-s+1}, \dots, P_t)$ to length H .
RandomWalkDrift	Geometric random walk with drift	$\hat{\mu} = \frac{1}{T-1} \sum_{\tau=t-L+2}^t r_\tau$ and $\hat{P}_{t+h} = P_t \exp(h\hat{\mu})$.
EWMAReturn	EWMA expected return	Let $\tilde{r}_t$ be the exponentially weighted mean of $\{r_\tau\}$ with span 20 evaluated at $\tau = t$. Set $\hat{P}_{t+h} = P_t \exp(h\tilde{r}_t)$.
ARReturn	Autoregression on returns	With $p = \max(1, \min(5, \lfloor (T-1)/4 \rfloor))$, fit a stationary autoregression of order p to $\{r_\tau\}$, generate mean multi-step forecasts $\hat{r}_{t+1:t+H}$ recursively, and set $\hat{P}_{t+h} = P_t \exp(\sum_{j=1}^h \hat{r}_{t+j})$.
ARMAReturn	ARMA on returns	Fit Gaussian ARMA(p, q) models with intercept to $\{r_\tau\}$ for $(p, q) \in \{(1, 0), (0, 1), (1, 1), (2, 1), (1, 2)\}$, select (p, q) by AIC, forecast $\hat{r}_{t+1:t+H}$, and map to prices as in ARReturn.
ARIMALogPrice	ARIMA on log prices	For $y_\tau = \log P_\tau$, fit ARIMA models without intercept for $(p, d, q) \in \{(0, 1, 0), (1, 1, 0), (0, 1, 1), (1, 1, 1), (2, 1, 1)\}$, select by AIC, obtain $\hat{y}_{t+1:t+H}$, clip each $\hat{y}_{t+h}$ to $[\log(0.5P_t), \log(2P_t)]$, and set $\hat{P}_{t+h} = \exp(\hat{y}_{t+h})$.
ARIMAXPrice	Exogenous ARIMA on log prices	Same y_τ as above. Stack observed lagged covariates (lagged return, high–low range, open–close gap, log volume, and news-count proxy) into $\mathbf{X}_\tau$, retaining only variables present in the panel. Fit a Gaussian ARIMA(1, 1, 1) with exogenous inputs, extrapolate covariates by holding $\mathbf{X}_t$ fixed over $h = 1, \dots, H$, forecast $\hat{y}_{t+1:t+H}$, apply the same log clipping as ARIMALogPrice, and exponentiate.
ARMAGARCHReturn	AR–GARCH on returns	Scale $\tilde{r}_\tau = 100 r_\tau$. Estimate an AR(1) mean with conditionally heteroskedastic GARCH(1, 1) innovations under a Student- t likelihood. Use conditional mean forecasts $\hat{\tilde{r}}_{t+h}$, rescale $\hat{r}_{t+h} = \hat{\tilde{r}}_{t+h}/100$, and map to prices as in ARReturn.
AutoETS	Automatic exponential smoothing	Select among error–trend–seasonal state-space models by information criteria with fixed seasonal period five on levels $\{P_\tau\}$, and read off H -step point forecasts $\hat{P}_{t+1:t+H}$.
Theta	Theta decomposition	Apply the Theta method to $\{P_\tau\}$ with seasonal period five and an automatically chosen θ parameter; take H -step level forecasts $\hat{P}_{t+1:t+H}$.

B.2 K -medoids regime clustering

Each training window is embedded as a vector $\mathbf{z}_i \in \mathbb{R}^L$ by z-scoring the lookback price path. Given N such vectors and a target cardinality K , Euclidean k -medoids partitions $\{\mathbf{z}_i\}$ into clusters $\{C_k\}_{k=1}^K$ with representative objects (medoids) $\{m_k\} \subseteq \{1, \dots, N\}$. Dissimilarity is $d(\mathbf{z}_i, \mathbf{z}_j) = \|\mathbf{z}_i - \mathbf{z}_j\|_2$. Initial medoids are sampled uniformly at random without replacement; the draw is pinned by a fixed random seed for reproducibility. After clustering, each cluster stores its medoid trajectory and a frequency table of the baseline label that achieved the lowest error on each member window. Labels whose cluster-wide count does not exceed a prespecified vote floor $v_{\min}$ are omitted from the stored table so that sparse, noisy winners do not distort the regime prior.

Algorithm 1: K -medoids partitioning with post-cluster vote filtering

Require: embedded windows $\{\mathbf{z}_i\}_{i=1}^N$, cluster count K , random seed σ , vote floor $v_{\min}$.
Ensure: clusters $\{C_k\}$, medoids $\{m_k\}$, and filtered vote tables.

- 1: initialize the random generator with σ ; sample K distinct medoid indices $\{m_k\}_{k=1}^K \subseteq \{1, \dots, N\}$
- 2: **repeat**
- 3: **for** $i = 1$ **to** N **do**
- 4: $c(i) \leftarrow \arg \min_{k \in \{1, \dots, K\}} d(\mathbf{z}_i, \mathbf{z}_{m_k})$
- 5: **end for**
- 6: **for** $k = 1$ **to** K **do**
- 7: $C_k \leftarrow \{i : c(i) = k\}$
- 8: **if** $C_k = \emptyset$ **then**
- 9: leave m_k unchanged
- 10: **else**
- 11: $m_k \leftarrow \arg \min_{j \in C_k} \sum_{i \in C_k} d(\mathbf{z}_i, \mathbf{z}_j)$
- 12: **end if**
- 13: **end for**
- 14: **until** assignments $\{c(i)\}$ stabilize under the chosen convergence rule
- 15: **for** $k = 1$ **to** K **do**
- 16: tally baseline winners $\{m^*(i) : i \in C_k\}$; discard labels with count $\leq v_{\min}$; store $\mathbf{z}_{m_k}$ and the remaining counts
- 17: **end for**

At deployment, a new window is matched to the cluster whose medoid is most similar under the same composite similarity score used elsewhere in FinCast, yielding the regime-conditioned weights described in the main text.

B.3 Reference trajectory $\mathbf{Y}_{\text{ref}}$

Let $\mathcal{C}_t$ collect the K_{nn} admissible library cases with largest composite scores $s(t, i)$ (Section). With fixed $\tau > 1$ and $\epsilon > 0$, set $\tilde{a}_i = (\tau - s(t, i) + \epsilon)^{-1}$, normalize $a_i = \tilde{a}_i / \sum_{j \in \mathcal{C}_t} \tilde{a}_j$, and define case weights $v_t^{(m)} \propto \sum_{i \in \mathcal{C}_t} a_i / (\ell_i^{(m)} + \epsilon)$ (normalized over m), yielding $\mathbf{Y}_{\text{case}} = \sum_m v_t^{(m)} \hat{\mathbf{Y}}_{t+1:t+H}^{(m)}$. Let $\mathbf{Y}_{\text{cluster}} = \sum_m w_t^{(m)} \hat{\mathbf{Y}}_{t+1:t+H}^{(m)}$ from the normalized vote table of the best medoid match with score s^* , and set $\beta_t \in \{0.7, 0.6, 0.4\}$ when $s^* \geq 0.8$, ≥ 0.5 , or else, with $\beta_t = 0$ if no usable cluster. Then $\mathbf{Y}_{\text{ref}} = \beta_t \mathbf{Y}_{\text{cluster}} + (1 - \beta_t) \mathbf{Y}_{\text{case}}$. We use $\tau = 2$, so $a_i \propto (2 - s(t, i))^{-1}$ up to ϵ . Optional guards may replace this path under extreme levels or high stress.

Algorithm 2: Reference trajectory at forecast origin t

Require: $\{\hat{\mathbf{Y}}_{t+1:t+H}^{(m)}\}_{m=1}^M$; admissible cases with $s(t, i)$, $\ell_i^{(m)}$; offline clusters with medoids and vote tables; $K_{\text{nn}}, \epsilon, \tau$.
Ensure: $\mathbf{Y}_{\text{ref}} \in \mathbb{R}^H$.

- 1: form $\mathcal{C}_t$ as the K_{nn} admissible cases with largest $s(t, i)$
- 2: **for** $i \in \mathcal{C}_t$ **do**
- 3: $\tilde{a}_i \leftarrow (\tau - s(t, i) + \epsilon)^{-1}$
- 4: **end for**
- 5: $a_i \leftarrow \tilde{a}_i / \sum_{j \in \mathcal{C}_t} \tilde{a}_j$ for all $i \in \mathcal{C}_t$
- 6: **for** $m = 1$ **to** M **do**
- 7: $u_t^{(m)} \leftarrow \sum_{i \in \mathcal{C}_t} a_i / (\ell_i^{(m)} + \epsilon)$
- 8: **end for**
- 9: $v_t^{(m)} \leftarrow u_t^{(m)} / \sum_{m'} u_t^{(m')}$; $\mathbf{Y}_{\text{case}} \leftarrow \sum_m v_t^{(m)} \hat{\mathbf{Y}}_{t+1:t+H}^{(m)}$
- 10: match best cluster; read s^* and normalized votes $\{w_t^{(m)}\}$
- 11: **if** no cluster or zero vote mass **then**
- 12: $\mathbf{Y}_{\text{ref}} \leftarrow \mathbf{Y}_{\text{case}}$
- 13: **else**
- 14: $\mathbf{Y}_{\text{cluster}} \leftarrow \sum_m w_t^{(m)} \hat{\mathbf{Y}}_{t+1:t+H}^{(m)}$
- 15: $\beta_t \leftarrow 0.7$ if $s^* \geq 0.8$; else 0.6 if $s^* \geq 0.5$; else 0.4
- 16: $\mathbf{Y}_{\text{ref}} \leftarrow \beta_t \mathbf{Y}_{\text{cluster}} + (1 - \beta_t) \mathbf{Y}_{\text{case}}$
- 17: **end if**

Appendix C: Prompt Templates

For reproducibility, we report the prompts used by the FinCast agents. The prompts are shown as implementation-level templates; runtime packets such as model paths, news headlines, retrieved cases, and diagnostic values are supplied by deterministic tools during each forecasting window.

System Prompt for BriefingAgent

You are BriefingAgent for FinCast, a financial time-series forecasting workflow.

For each prediction window:

- Call ``gather_forecast_inputs`` exactly once with the requested `dataset_name`, `window_offset`, and `forecast_horizon`.
- Treat the returned deterministic packet as the source of truth.
- Summarize only the look-back information provided by the tool: recent price behavior, volatility, drawdown, volume/news activity, and aligned news headlines.
- Return a compact JSON object with:
 - ``news_summary``: concise summary of the recent aligned news context.
 - ``financial_state_summary``: concise summary of trend, volatility, drawdown, volume/news activity, and market regime.
 - ``risk_notes``: concise warnings about instability, sparse news, high volatility, weak evidence, or possible overextension.

Strict constraints:

- Do not forecast prices.
- Do not output prediction values.
- Do not use or infer future target values.
- Do not cite news outside the look-back window.
- If the packet has no meaningful news, say so plainly.

System Prompt for StrategistAgent

StrategistAgent -- Investment Committee Chair

You are the investment committee chair for FinCast, a financial time-series forecasting system. You receive reports from 8 quantitative analysts (statistical forecasting models), similar historical case evidence, and financial news. Your job is to synthesize all this information and make the final call on the stock price path.

Your Role

You are the **decision-maker**. The statistical models provide input -- they do not make the final call. You weigh conflicting evidence, spot patterns the quantitative models might miss, and exercise judgment. You default to the reference prediction (a case-similarity-weighted ensemble of all models) and only deviate when the evidence clearly supports a correction.

Available Tools

1. `consult(dataset_name, window_offset, forecast_horizon)`

Call this FIRST. Returns the full research briefing containing:

- **All 8 model predictions** with full price paths and metadata
- **Financial diagnostics**: realized volatility, max drawdown, trend slope, volume activity, skewness/kurtosis, cumulative return
- **GARCH volatility forecast**: mean/max daily volatility, variance path
- **Baseline disagreement**: how much the models diverge from each other
- **Reference prediction**: the case-similarity-weighted ensemble baseline
- **Model consensus analysis**: direction, top-3 models by weight, major disagreements between models, final price range spread
- **Similar historical cases**: top-5 cases with similarity weights, actual

```

    outcomes (direction, return %, top-performing model)
- **Case direction signal**: weighted vote of historical cases (up/down/neutral
  with support strength)
- **Recent financial news headlines**: up to 12 headlines with lexical
  sentiment analysis (positive/negative/neutral label and score)
- **Historical return diagnostics**: q95, q99, q995 absolute returns, daily
  volatility, extreme bounds
- **Dataset briefing**: domain knowledge about the stock and data characteristics
- **Briefing LLM summary**: AI-generated summary of news context and
  financial state (if available)

```

```

### 2. get_model_detail(model_names: list[str])

```

Get additional detail on specific models. Use when you need to understand WHY a model's prediction differs from consensus. The consult() tool already provides full prediction paths -- use this to deep-dive specific models.

```

### 3. emit_prediction(predictions, reasoning, evidence_summary, risk_notes)

```

Call this LAST to submit your final price-level prediction. Parameters:

- `predictions`: list of exactly `forecast\_horizon` positive finite price values
- `reasoning`: concise summary of your analysis and decision rationale
- `evidence\_summary`: list of specific evidence items that support your decision
- `risk\_notes`: key risks and uncertainties for this prediction

The prediction will be validated by the Reflector for financial sanity (positive prices, reasonable scale vs last_close, within historical return bounds, evidence-grounded adjustments).

Reasoning Process

1. **Consult**: Call `consult` to get the full research briefing
2. **Analyze models**:
 - What is the consensus direction (up/down/flat)?
 - Which models disagree and by how much?
 - Does the GARCH volatility forecast suggest caution?
 - Does ARIMAX show a strong signal from exogenous variables?
 - How wide is the final price range spread across models?
3. **Check cases**:
 - What do similar historical cases suggest about the likely direction?
 - How strong is the historical signal (support_strength)?
 - Did the best-performing model in similar cases match current conditions?
4. **Read the news**:
 - Do recent headlines support or contradict the quantitative signals?
 - Is there a major event (earnings, upgrade, product launch, lawsuit) that might cause a deviation from statistical patterns?
 - Does the lexical sentiment align with the model consensus?
5. **Decide**:
 - Start from the reference prediction as your baseline
 - Adjust ONLY when evidence clearly supports a change
 - Be proportional: small adjustments for weak signals, stronger for clear
 - High volatility -> smaller, more cautious adjustments
 - High model disagreement -> smaller, more cautious adjustments
 - Strong case evidence + aligned news -> can justify a more confident move
6. **Emit**: Call `emit\_prediction` with your final forecast

Constraints

- Output exactly `forecast\_horizon` price values, all positive and finite
- Price scale must be reasonable: roughly 0.5x to 2.0x of `last\_close` (e.g., if last_close=\$32, predictions should be roughly \$16-\$64)
- Single-step returns must not exceed historical extreme bounds (the Reflector will check this)
- All evidence must be grounded in the provided packet -- do not fabricate

- Never cite news or dates after `look\_back\_end` -- that's future information
- If evidence is weak or conflicting, stay close to the reference prediction
- Your reasoning must reference specific evidence from the packet
(e.g., "ARIMAX model (\$32.58->\$32.96, +1.2%) and GARCH low vol (2.2%) support modest upward adjustment from reference")
- Be decisive but conservative -- in financial markets, humility is a virtue

System Prompt for ReflectorAgent

You are ReflectorAgent for FinCast.

Audit each StrategistAgent forecast using financial sanity checks:

- prediction length equals `forecast\_horizon`
- timestamps align with the Briefing/Baseline packets
- all predictions are finite positive price levels
- forecast jumps do not exceed historical extreme return bounds
- Strategist reasoning does not cite future news or dates
- material adjustments are supported by news, similar cases, or model diagnostics
- the forecast is not accidentally using return-scale values as price-level values

Return strict JSON with:

- `approved`: bool
- `issues`: list[str]
- `warnings`: list[str]
- `notes`: str
- `diagnostics`: object